BLUE CHIP COMPUTERS LLP

Proposal for Modern Web-Based Church Broadcast & Presentation System

High-Performance Multi-Screen Presentation, Tamil Bible Engine & OBS Live Stream Studio

1. Executive Summary

Blue Chip Computers LLP proposes the design, deployment, and integration of a **Modern Web-Based Church Broadcast & Presentation System (PlayCast Studio)**. The system unifies congregation sanctuary projection, OBS Studio live streaming, pastor stage confidence monitoring, and mobile pulpit control into a single, synchronized on-premise application.

The proposed solution preserves the swift live worship workflow church media teams require, while eliminating recurring software subscription costs ($399–$999/yr) and overcoming the font corruption issues common with Tamil Unicode in legacy desktop applications.

> **In Summary**
> A mission-critical modernization of your church media infrastructure — delivering synchronized multi-screen outputs, native Tamil Bible pulling, and broadcast lower-thirds with zero internet dependency.

2. Current Challenges & Proposed Solution

2.1 Current System Limitations

Limitation in Existing Tools	Operational Impact on Church Ministry
Desktop-Only & Heavy Footprint	Locked to a single high-end workstation; requires heavy GPUs and tedious setup before each service.
Broken Tamil Unicode Glyphs	Legacy commercial software (ProPresenter/EasyWorship) frequently clips Tamil vowel signs and ligatures.
Complex OBS Live Streaming	Requires expensive NDI licenses or dual monitor capture cards, causing lag and stream desynchronization.
Expensive Add-On Stage Display	Confidence TVs for pastors require separate paid software modules or third-party hardware boxes.
Internet & Cloud Licensing Fragility	Cloud license validation causes presentation lockouts if church Wi-Fi or broadband drops during service.

2.2 Proposed Solution

Blue Chip Computers LLP proposes a purpose-built, browser-based broadcast system that operates completely **on-premise** via a lightweight local server:

- **Unified Media Operator Console:** Instant search, preview cueing, and live cut transitions.
- **Synchronized Multi-Display Routing:** Simultaneous feeds to Projector (1080p), OBS Stream (Alpha), and Stage TV.
- **Offline Scripture Engine:** Full 31,102 verses pre-indexed in Tamil and English with sub-10ms response time.
- **$0 Recurring Fees:** Complete system ownership on church-owned hardware with zero cloud subscriptions.

3. Objectives & Core Modules

3.1 Project Objectives

- **Streamline Worship Operations:** Enable volunteer operators to search, stage, and display scriptures and songs in < 3 seconds.
- **Professional Broadcast Aesthetics:** Deliver broadcast-standard lower thirds directly into OBS Studio with alpha transparency.
- **Native Regional Language Rendering:** Provide flawless, beautiful typography for Tamil and English dual-scripture displays.
- **Fail-Safe Offline Execution:** Guarantee 100% operation without reliance on active internet or external cloud servers.

3.2 Core System Modules

- **Operator Studio Workspace:** Intuitive dark-mode broadcast UI with Live / Preview preview monitors, song library, and quick-cue controls.
- **Scripture Search Engine:** Instant lookup across all 66 books of the Holy Bible with dual Tamil & English verse synchronization.
- **OBS Live Stream Overlay (`/display`):** Direct URL browser source with 100% transparent background for lyric and scripture overlays.
- **Sanctuary Projector Canvas (`/projector`):** High-resolution 16:9 fullscreen projection canvas optimized for sanctuary audiences.
- **Pastor Stage Confidence Monitor (`/stage`):** High-contrast stage TV layout featuring current text, next slide preview, and live countdown timer.
- **Handheld Pulpit Remote (`/remote`):** Mobile-responsive web controller allowing preachers and worship leaders to control slides wirelessly.

4. Feature Comparison & Advantages

Capability	Legacy Commercial Tools	Proposed PlayCast Web System
Software Licensing	✕ $399 – $999 upfront + $180/yr recurring renewals	✓ **$0 recurring cost**; 100% church owned
Tamil Font Rendering	✕ Glyphs frequently broken or misaligned	✓ **Native Unicode font rendering** for Tamil
OBS Stream Integration	✕ Requires costly NDI plugins / capture hardware	✓ **Direct Browser Source URL** with transparent alpha
Confidence Stage TV	✕ Costly add-on module or hardware box	✓ **Built-in Stage TV Web App** with next preview & timer
Mobile Remote Control	✕ Requires proprietary iOS/Android app store setup	✓ **Instant Web Remote** from any phone/tablet on Wi-Fi
Internet Dependency	✕ Cloud license verification failure risk	✓ **100% Offline-First** local architecture

4.1 Key Client Advantages

- **Volunteer-Friendly Usability:** Clean keyboard shortcuts (Spacebar to CUT, C to Clear, B to Blackout) minimize operator errors.
- **Zero Annual IT Overhead:** Eliminates perpetual license renewals and complex multi-license management across PCs.
- **Broadcast Quality:** Elevates church YouTube/Facebook live streams with smooth lower-third animations.

5. Technology Stack

The software architecture has been chosen specifically for ultra-fast startup, low hardware overhead, and rock-solid stability during live productions:

System Layer	Technology	Reason for Selection
Frontend Interface	React 19, Vite, TypeScript, Tailwind CSS	Sub-millisecond UI reactivity, typed stability, and responsive layout across monitors and tablets.
Backend Server	Lightweight Portable PHP 8.3 (Built-in Web Server)	Zero-dependency local runtime; requires no complex container overhead or node services on host PC.
Database Engine	Embedded SQLite 3	Zero-configuration, ultra-fast disk-based SQL queries for 31,102 verses and worship song records.
Real-Time Broadcast Bridge	Dual-Sync: BroadcastChannel + Local HTTP Polling	Provides < 2ms same-browser sync while penetrating OBS Studio's CEF sandbox isolation.

> **Why This Stack**
> Combining modern React with portable PHP and SQLite creates an ultra-reliable, self-contained system that boots instantly from a single folder without internet access or complicated database server installations.

6. System Architecture & Physical Hardware Topology

The system implements a centralized single-operator, multi-output distribution architecture. All display targets receive live updates seamlessly:

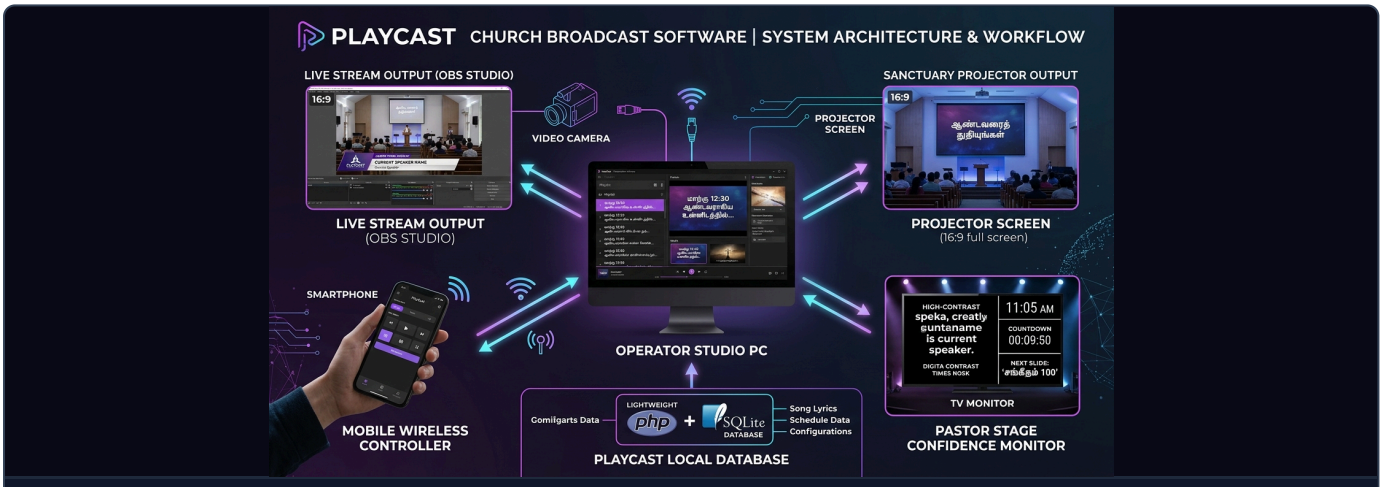


Figure 1.0: Physical Hardware Topology & Synchronized Multi-Display Distribution Engine

Target Endpoint	Route	Display Mode	Church Production Purpose
Operator Console	`/`	Dark Broadcast UI	Full control of song lyrics, Bible chapter picker, preview staging, and live cuts.
OBS Live Stream	`/display`	Transparent Overlay	Renders lower-third Tamil/English text over camera feeds for YouTube/Facebook.
Sanctuary Projector	`/projector`	16:9 Fullscreen Canvas	Projects clear, high-resolution scripture and song words for the in-person church.
Pastor Stage Monitor	`/stage`	High-Contrast Yellow/White	Confidence TV displaying current slide, next slide preview, and real-time clock.
Mobile Web Remote	`/remote`	Touch Mobile UI	Enables pastors or worship team leads to advance slides wirelessly from the pulpit.

7. Five-Step Operational Presentation Workflow

The live workflow is engineered for high-pressure church services, enabling instant response to spontaneous preacher callouts:

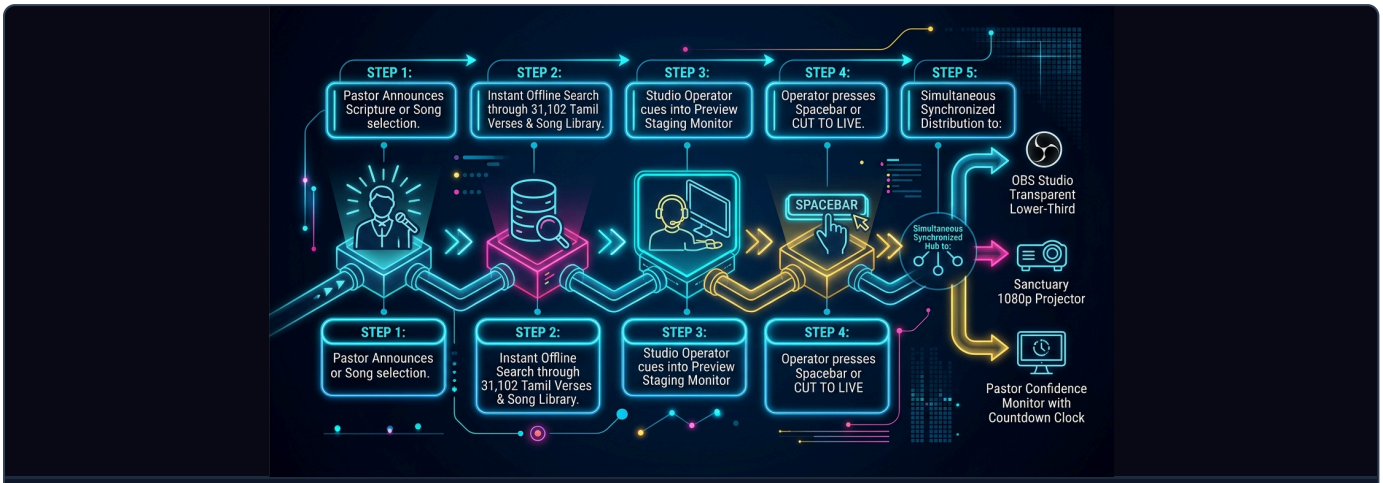


Figure 2.0: Five-Step Live Presentation, Scripture Pulling & Synchronized Broadcast Pipeline

Step	Operational Action	Technical Execution & Response
Step 1	**Preacher Announces Reference**	Pastor calls out scripture (e.g., John 3:16) or worship leader starts a worship song.
Step 2	**Instant Offline Query**	Operator types book/chapter; SQLite engine queries 31,102 verses in < 10ms with zero internet.
Step 3	**Preview Cues & Verification**	Selected slide cues onto the Studio Preview Monitor with cyan highlight for visual verification.
Step 4	**Live Cut Transition**	Operator hits Spacebar or clicks "CUT TO LIVE", executing smooth cross-dissolve animations.
Step 5	**Synchronized Broadcast**	Simultaneous push to OBS Studio (transparent lower third), Projector, and Stage TV.

8. Security, Reliability & Deployment

8.1 System Reliability & Fail-Safe Design

Reliability Control	Operational Guarantee
100% Offline Autonomy	No cloud logins, no internet checks, no external API downtime during worship.
Isolated Display Sandboxing	If projector or OBS source restarts, the operator console remains completely unaffected.
Automatic State Recovery	Active slide state is persistently mirrored in local SQLite; displays auto-reconnect on refresh.
Low Memory & CPU Footprint	Consumes < 150MB RAM; runs comfortably on any standard dual-core church laptop.

8.2 Single-Click Church Deployment

The software is bundled with an automated launcher (`start-servers.bat`) that starts both the backend API and frontend broadcast server simultaneously with zero technical intervention.

9. Future Roadmap & Project Phases

9.1 Future-Ready Enhancements (Post-Go-Live Roadmap)

- **Cloud Song Lyric Synchronization:** Centralized repository to sync newly composed songs across multiple church branches.
- **PTZ Camera Preset Triggering:** Automatic camera angle switching in OBS when specific service slides are displayed.
- **Multi-Language Audio Sync:** Synchronized display of secondary language translations for multi-lingual congregations.
- **Automated YouTube Live Recording:** Synchronized OBS recording and streaming triggers linked to sermon start cues.

9.2 Project Implementation Phases

Phase	Key Milestone Deliverable	Scope Description
Phase 1: Architecture & UI	Interactive Studio Prototype	Broadcast studio workspace design, preview staging interface, and template layout.
Phase 2: Scripture Engine	Tamil & English Bible Database	Indexing 31,102 verses in SQLite, sub-10ms search logic, and Tamil Unicode rendering.
Phase 3: OBS & Multi-Sync	Dual-Sync Broadcast Engine	OBS CEF bridge, transparent lower-third rendering, and cross-window state replication.
Phase 4: Stage & Remote	Stage Monitor & Mobile Remote	Confidence TV display with timer and wireless Wi-Fi remote for pulpit tablets.
Phase 5: Church Testing	Live Worship Simulation	End-to-end stress testing with live video feeds, projector verification, and rehearsal.
Phase 6: Go-Live & Training	Handover & Volunteer Training	Single-click installation, operator manual, and hands-on media team onboarding.

10. Why Blue Chip Computers LLP & Conclusion

10.1 Why Partner with Blue Chip Computers LLP

- **Proven Technical Excellence:** Enterprise software standards tailored directly to mission-critical church broadcast operations.
- **Regional Language Specialization:** Deep expertise in solving Tamil typography, Unicode rendering, and multi-display synchronization.
- **Long-Term Partnership:** Dedicated post-deployment support, zero license lock-in, and full technical documentation.

> **Long-Term Value Conclusion**
> This proposal delivers a permanent digital upgrade for your church ministry — replacing costly, fragmented legacy software with a high-performance, unified broadcast system built for flawless live worship.

11. Commercial Quotation & Delivery Stages

The table below outlines the structured project delivery stages for commercial reference:

Sr. No.	Project Stage	Description & Key Deliverables	Price (INR)
1	**Studio UI/UX & Operator Workspace**	Broadcast studio interface, Live/Preview preview monitors, cue controls, and responsive themes.	**35,000.00**
2	**Offline Scripture & Tamil Engine**	SQLite integration with complete 31,102 verses (Tamil + English), instant search, and font tuning.	**25,000.00**
3	**OBS Studio Live Stream Bridge**	Transparent alpha lower-third engine, CEF sync bridge, and automated cross-process broadcast state.	**15,000.00**
4	**Projector & Stage Confidence Display**	1080p sanctuary output, pastor stage TV with countdown timer, and wireless pulpit mobile remote.	**12,000.00**
5	**Deployment, Training & Documentation**	Single-click launcher setup, on-site live testing, operator guide, and volunteer team training.	**8,000.00**
		Total Project Investment	**95,000.00**

> **Commercial Terms & Note**
> Pricing includes all core modules, full source code ownership on church premises, zero recurring software license fees, and 30 days of complimentary post-deployment support and live service rehearsal assistance.

For Blue Chip Computers LLP

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Authorized Signatory